LAB # 01

EXERCISE:

Q1. Name the stages involved in the software testing life cycle, differentiate test planning and test analysis.

1. Stages in the Software Testing Life Cycle

The stages involved in the Software Testing Life Cycle are:

- Test Planning
- Test Analysis
- Test Design
- Construction and Verification
- Testing Cycles
- Final Testing and Implementation
- Post-Implementation

Q2. Differentiate between white box and black box testing techniques. Which technique do you prefer where you are asked to perform testing at component level and why? Explain with suitable examples

Difference Between Test Planning and Test Analysis:

Black Box Testing	White Box Testing
Requires deep knowledge of internal code structure.	Does not require knowledge of internal code.
Performed by developers.	Performed by QA testers or end users.
Unit and integration testing.	System and user acceptance testing.
Testing a function to ensure it processes correctly.	Testing login functionality without knowing code details.

Technique at Component Level

White Box Testing is preferred for component/unit testing because it ensures individual units function correctly based on internal code structure.

Example: White box testing for calculateSum(x, y) ensures all possible combinations of x and y are tested to verify the internal logic, covering conditions like positive, negative, zero, and boundary values.

Q3. Explain software testing levels .

Software Testing Levels

1. **Unit Testing**: Focuses on testing the smallest parts of software (functions, methods). Performed by developers.
2. **Integration Testing**: Verifies the interaction between modules or units. Can use bottom-up or top-down approaches.
3. **System Testing**: Tests the entire system as a whole to ensure compliance with requirements. Performed by QA teams.
4. **User Acceptance Testing (UAT)**: Validates the system against user requirements and determines readiness for release. Involves end-users.

Q4. Write down some advantages of software testing.

Advantages of Software Testing

- Software testing identifies and prevents defects early, saving cost and time.
- Software testing ensures software meets specified requirements and quality standards to improve software quality.
- Software testing ensures that the build software meets customer requirements enhancing their satisfaction
- Software testing reduces risks by identifying errors, failures ensuring system stability.
- Software testing helps making software easier to maintain so that it can updated without any difficulties.

Q5. Write a test set for copying a file XYZ from folder A to folder B. (at least 5 test cases).

TEST CASES

1 To check whether folder A exists in drive

2 To check whether folder B exists in drive

3 To check whether File XYZ exists in folder A

4 To check whether Folder B has sufficient space

5 To check whether File XYZ can be copied from folder A

6 To check whether File XYZ can be pasted in folder B

7 To check whether file XYZ already exists in folder B.